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Delineating the barotropic and baroclinic mechanisms in the midlatitude eddy-driven jet response to lower-tropospheric thermal forcing

Yu Nie, Yang Zhang*

Institute for Climate and Global Change Research, School of Atmospheric Sciences, Nanjing University, Nanjing, China

Gang Chen

Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, NY, USA

Xiu-Qun Yang

Institute for Climate and Global Change Research, School of Atmospheric Sciences, Nanjing University, Nanjing, China

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**Corresponding author address:* Yang Zhang, 22 Hankou Road, School of Atmospheric Sciences, Nanjing University, Nanjing, China

E-mail: yangzh@alum.mit.edu

ABSTRACT

Observations and climate models have shown that the midlatitude eddy-driven jet can exhibit an evident latitudinal shift in response to lower-tropospheric thermal forcing (e.g., the tropical SST warming during El Nino or extratropical SST anomalies associated with the atmosphere-ocean-sea ice coupling). In addition to the direct thermal wind response, the eddy feedbacks, including baroclinic mechanisms such as the lower-level baroclinic eddy generation and barotropic mechanisms such as the upper-level wave propagation and breaking, all can contribute to the atmospheric circulation response to lower-level thermal forcing, but their individual roles have not been well explained. In this study, using a nonlinear β-plane multi-level quasi-geostrophic channel model, the mechanisms through which the lower-level thermal forcing induces the jet shift are investigated. By diagnosing the finite amplitude wave activity budget, the baroclinic and barotropic eddy feedbacks to the lower-level thermal forcing are delineated. Particularly, by examining the transient circulation response after thermal forcing is switched on, it is shown that the lower-level thermal forcing affects the eddy-driven jet rapidly by modifying the upper-level zonal thermal wind distribution and the associated meridional wave propagation and breaking. The anomalous baroclinic eddy generation, however, acts to enhance the latitudinal shift of the eddy-driven jet only in the later stage of transient response. Furthermore, the barotropic mechanism is explicated by overriding experiments in which the barotropic flow in the vorticity advection is prescribed. Unlike the conventional baroclinic view, the barotropic eddy feedback, particularly the irreversible PV mixing through barotropic vorticity advection and deformation, plays a major role in the atmospheric circulation response to the lower-level thermal forcing.

1. Introduction

Understanding the dynamical mechanisms of the midlatitude atmospheric circulation response to the lower boundary thermal forcing is important for the seasonal and decadal prediction of midlatitude weather and associated extreme events. Observations and climate models have shown that the midlatitude eddy-driven jet exhibits an evident meridional displacement in response to the lower boundary thermal forcing, such as the tropical SST warming during El Nino (Seager et al. 2003; Lu et al. 2008; Chen et al. 2008), extratropical SST anomalies associated with the low-frequency variability in air-sea interactions (Kushnir et al. 2002; Brayshaw et al. 2008; Nakamura and Yamane 2009, 2010; Ogawa et al. 2012), and the recent Arctic warming trend (see the review in Cohen et al. (2014)). Collectively, these studies suggest that the local surface warming can alter the spatial distribution of the horizontal temperature gradient, resulting in a latitudinal shift of the lower-level baroclinicity that is responsible for the latitudinal shift of the midlatitude jet and storm tracks.

The extratropical atmospheric circulation response to a lower-level thermal forcing can be understood as a direct thermal wind response plus an indirect response due to eddy feedbacks (Kushner et al. 2001; Deser et al. 2004; Lu et al. 2013). The direct response of the jet stream often displays a baroclinic vertical structure (Hoskins and Karoly 1981; Kushnir et al. 2002). Following the latitudinal shift of the lower-level baroclinicity, the upper-level thermal wind would manifest a dipolar meridional structure, as illustrated in Fig.1a, indicating a latitudinal shift of the upper-level jet. In contrast, the indirect response of the midlatitude jet due to the eddy feedbacks, as argued in Deser et al. (2004) and Ring and Plumb (2007, 2008), often exhibits an equivalent barotropic vertical structure, with a magnitude larger than the direct thermal wind response. Moreover, the eddy feedback mechanism can be either baroclinic or barotropic. On the one hand, the lower-level

baroclinicity is generally considered as a key ingredient in the storm track dynamics (Hoskins and Valdes 1990; Held and O'Brien 1992). With a latitudinal shift in the lower-level baroclinicity, as illustrated in Fig.1b, more/less baroclinic eddies would be generated where the lower-level baroclinicity is enhanced/reduced, which, through upward eddy activity flux, can result in a latitudinal shift of eddy momentum flux convergence and the eddy-driven jet (Robinson 2000; Lorenz and Hartmann 2001; Chen and Plumb 2009; Zhang et al. 2012; Zurita-Gotor et al. 2014). On the other hand, the anomalous zonal wind in the upper troposphere (due to the thermal wind adjustment or the baroclinic eddy feedback), as shown in Fig.1c, may alter either the linear meridional wave propagation and critical layer absorption in the upper troposphere (Chen et al. 2008; Chen and Zurita-Gotor 2008; Harnik et al. 2010; Barnes 2010) or the nonlinear wave breaking and irreversible PV mixing (Riviere 2009; Wang and Magnusdottir 2011; Chen et al. 2013; Lu et al. 2013; Nie et al. 2014), which, in turn, would alter the eddy momentum flux and act to enhance the latitudinal shift of the eddy-driven jet.

The goal of the paper is to compare the baroclinic and barotropic mechanisms quantitatively in the atmospheric response to thermal forcing. Since most of the aforementioned mechanisms include the eddy-zonal flow interactions that are difficult to disentangle in the observations, we use a nonlinear β-plane multi-level quasigeostropic channel model. Moreover, we adopt a Finite Amplitude Wave Activity (FAWA, hereafter) diagnostic introduced by Nakamura and Zhu (2010), which can be easily implemented to evaluate the eddy-zonal flow interactions even in the presence of nonlinear wave breaking (Solomon et al. 2012; Chen et al. 2013; Sun et al. 2013). This formalism can attribute the eddy momentum forcing to the net effect of baroclinic eddy generation and irreversible PV mixing, which facilitates a separation of the baroclinic and barotropic eddy feedbacks in the jet shift to the low-level thermal forcing. By examining the transient circulation response after forcing is switched on, we found that the lower-level thermal forcing affects the jet

stream rapidly by altering the upper-level thermal wind and irreversible PV mixing. In contrast, the baroclinic eddy generation acts to enhance the jet shift only in the later stage of the response.

In addition, the barotropic eddy feedback, particularly the irreversible PV mixing, is explicated by overriding experiments in which the barotropic flow in the vorticity advection is prescribed. Although the above barotropic process is always coupled with the baroclinic eddy generation (Yu and Hartmann 1993) in a baroclinic eddy life cycle, Ferrari and Nikurashin (2010), using the advection-diffusion equation for passive tracers, showed that imposing a strong zonal jet suppresses the effective diffusivity in tracer transport in spite of the large baroclinic eddy generation and kinetic energy at the jet center. This motivates us to override the advecting flow in the PV equation analogous to modifying the zonal advecting flow in tracer transport. Moreover, this overriding technique decouples the barotropic advection and deformation of PV from the baroclinic energy conversion, thus allows us to quantify the changes in baroclinic energy conversion and barotropic irreversible PV mixing in response to the thermal forcing separately.

The structure of the paper is assigned as follows. Section 2 introduces the setup of the numerical experiments and the diagnostic methods used in the study. Equilibrated and transient responses of the numerical simulations are depicted in section 3, in which the dynamical processes responsible for the latitudinal shift of the eddy-driven jet are illustrated. The designs of overriding experiments are described in section 4. Section 5 shows the results of overriding experiments, through which the relative roles of barotropic and baroclinic eddy feedbacks in the jet shift in response to the lower-level thermal forcing are compared. Summary and discussion are presented in section 6.

2. Experiments Design and Analysis Methods

a. The β Plane Multi-Layer QG Model

A β plane multi-level Quasi-Geostrophic(QG) channel model with fixed static stability is used in this study to simulate the extratropical eddy-mean flow interactions in the Northern Hemisphere, similar to that of Solomon and Stone (2001) and Zhang et al. (2009, 2012). The model has a channel length of 21040 km, which is comparable with the latitudinal belt length in mid-latitude, and a channel width of 10000 km. The width of the baroclinic zone is set as half of the whole channel width. This model is symmetric about the mid-channel, thus the spherical geometry effect on the asymmetry of wave dynamics has been ignored and the model simulates a pure eddy-driven jet. The horizontal resolution is 156 km by 156 km, which can resolve the essential dynamics and eddy statistics (Zhang et al. 2009). To better simulate the eddy response to the vertical variation of the diabatic heating, the model is set with 17 equally spaced pressure levels.

In this model, the potential vorticity equation with diabatic heating and boundary layer friction is integrated:

$$\frac{\partial q}{\partial t} = -\mathbf{V}\cdot\nabla_h q - f_0\frac{\partial}{\partial p}\frac{HR}{spC_p} + \mathbf{k}\cdot\nabla\times\mathbf{Fr}, \tag{1}$$

where $q = \nabla^2\psi + \beta y + \frac{\partial}{\partial p}\frac{f_0^2}{s}\frac{\partial\psi}{\partial p}$ is the potential vorticity, $\mathbf{V} = (u_g, v_g)$ denotes the geostrophic winds. $R = 287Jkg^{-1}K^{-1}$ is the ideal gas constant, $C_p = 1004Jkg^{-1}K^{-1}$ is the specific heat of the air. The static stability parameter $s = -\frac{1}{\pi}\frac{\partial\overline{\theta}^{xy}}{\partial p}$, where $\overline{\theta}^{xy}$ denotes the horizontally averaged potential temperature and $\pi = \frac{p}{R}(\frac{p_0}{p})^{R/C_p}$, p_0 is the surface pressure. Fr denotes the frictional dissipation and the diabatic heating term H has two contributors: radiative convective heating H_r and thermal diffusion in the boundary layer H_d. In this study, the friction Fr and the boundary thermal diffusion H_d are parameterized as in Zhang et al. (2009), but with a simplified setting that confines them at the surface. Sensitivity studies have been carried out to guarantee that results in

this work do not depend on the setting of these two boundary layer forcing. Radiative convective heating H_r in this model is parameterized by the Newtonian cooling form:

$$H_r = C_p \frac{T_e - T}{\tau_r}, \tag{2}$$

where T_e is the atmospheric temperature in radiative-convective equilibrium (RCE) state. $\tau_r = 40$ days, is the relaxation time scale. Sensitivity studies also have been carried out to guarantee that the conclusions in this work are robust to the setting of this parameter, except that the transient response time to the radiative forcing will be faster with shortened τ_r. The meridional variation of the potential temperature of the RCE state in the troposphere is set below,

$$\theta_e^+(y,p) = \begin{cases} \frac{\Delta_e}{2} & -L/2 \leq y - y_0(p) < -L/4, \\ -\frac{\Delta_e}{2}\sin[\frac{\pi(y-y_0(p))}{L/2}] & -L/4 \leq y - y_0(p) \leq L/4, \\ -\frac{\Delta_e}{2} & L/4 < y - y_0(p) \leq L/2, \end{cases} \tag{3}$$

where $\theta_e^+ = \theta_e - \overline{\theta_e}^{xy}$, $\Delta_e = 43K$ is the temperature difference across the channel, L is the channel width. $y_0(p)$ is the center of the baroclinic zone at a given level p. In the stratosphere, the potential temperature gradient of the RCE state is set to one tenth of that in the tropopause and of opposite sign.

b. Setup of the Standard Experiments

In the standard experiments, the atmospheric response to the imposed thermal forcing is investigated by modifying the potential temperature in the RCE state. We start with a standard control (CTL) run simulation, in which $y_0(p) = 0$, thus the maximum upper and lower level temperature gradients are both at the channel center, as described by the RCE state $\partial T_e/\partial y$ in Fig.2a. We then carry out a set of standard LOW run experiments by systematically displacing the latitude of the maximum lower-level $\partial T_e/\partial y$ poleward and setting the center of the baroclinic zone $y_0(p)$ as

$$y_0(p) = \begin{cases} \frac{\Delta y}{2}(\cos(\pi \frac{p_0 - p}{p_0 - p_T}) + 1) & p_0 \leq p \leq p_T \\ 0 & p_T \leq p \leq 0, \end{cases} \tag{4}$$

in which p_T=312.5 hPa is the pressure of the tropopause and Δy denotes the displacement of the lower layer baroclinic zone from the mid-channel. As displayed in Fig.2b, in the LOW run experiments, the centre of the upper-level baroclinic zone is fixed, while the lower-level baroclinic zone is shifted poleward gradually from the mid-level down to the surface. In the standard LOW run experiments, four sensitivity runs are carried out by increasing Δy incrementally with a step of 625 km, which is Δy= 625 km, 1250 km, 1875 km , and 2500 km.

The atmospheric response in the above numerical experiments is investigated through equilibrated response analysis and transient evolution analysis. For the equilibrated analysis, each simulation starts from the RCE state with small amplitude perturbations added at the initial moment. Then the experiments are integrated for 11000 days, and the equilibrium statistics are computed over the last 10000 days. For the transient response analysis in the standard simulations, 200-member ensembles of perturbation experiments are carried out, in which the model starts from the equilibrium state of the standard CTL run and the thermal forcing is suddenly switched on. The transient experiments allow us to dissect the time evolution from the initial control state to the new forced equilibrium state and help understand the atmospheric response to the lower-level thermal forcing. To form the 200-member ensemble, the initial condition for each transient run is set by branching out from the last day of each 50-day segment of the 10000-day long equilibrium state in the standard CTL run. Then the same thermal perturbation is turned on instantaneously and the model is integrated forward 200 days to a new statistical equilibrium.

c. Analysis method: the Finite-Amplitude Wave Activity Diagnostics

To elucidate the relative importance of the barotropic and baroclinic processes in the atmospheric response to the lower-level thermal forcing, the FAWA diagnostics are adopted in our study. The hybrid Eulerian-Lagrangian FAWA diagnostics developed by Nakamura and Zhu (2010) facilitates a separation and quantification of barotropic processes such as wave propagation, breaking and resultant irreversible PV mixing, and baroclinic processes such as baroclinic eddy generation and vertical propagation in the eddy-mean flow interactions. The FAWA is defined as the waviness of the PV contours at a constant pressure level.

$$A(Y_e) = \frac{1}{L_{ength}}\left(\iint_{Q(Y_e)\leq q(y), y\leq Y_0} q dS - \iint_{Y_e\leq y\leq Y_0} q dS\right), \tag{5}$$

where y denotes latitude and L_{ength} is the length of the channel. Y_e is the equivalent latitude corresponding to the wavy contour $q = Q$, which is determined by the requirement that the area between the PV contour Q and the northern boundary Y_0 of the model equals the area bounded by Y_e and Y_0. An example of the diagnostic is illustrated in Fig.3a, which shows a snapshot of the upper-level PV on day 1300 in the standard CTL run. The solid contours denote the PV value corresponding to the equivalent latitudes highlighted with dashed lines. For the PV value $q = Q$, i.e. the PV contour highlighted in black, its wave activity is equal to the integral of PV over the region south of Y_e and north of Q (red region) minus the integral of PV south of Q and north of Y_e (blue region).

The eddy-driven jet, which is denoted by the vertically averaged zonal-mean zonal wind, is primarily forced by the vertically-averaged eddy momentum flux convergence and damped by the boundary layer friction. As the eddy momentum flux convergence peaks in the upper troposphere, whose contribution dominates its vertical integral (Nie et al. 2014), in this study we focus on

the response of the upper-level eddy momentum flux convergence to understand the jet response. Following the formalism in Nakamura and Zhu (2010) and Chen et al. (2013), the eddy momentum flux convergence satisfies

$$M \equiv -\frac{\partial([u^*v^*])}{\partial y} = \underbrace{-\frac{\partial}{\partial p}\left(\frac{f[v^*\theta^*]}{\partial\overline{\theta}^{xy}/\partial p}\right)}_{baroclinic} \underbrace{-k_{eff}\frac{\partial Q}{\partial Y_e}}_{barotropic} - \frac{\partial A}{\partial t} + \Delta\Sigma. \tag{6}$$

The physical meaning of each term on the right hand side of the equation is elaborated as follows:

- $-\frac{\partial}{\partial p}\left(\frac{f[v^*\theta^*]}{\partial\overline{\theta}^{xy}/\partial p}\right)$ is the vertical differential of poleward eddy heat flux divided by the static stability parameter (hereafter referred as baroclinic term), which represents the contribution from baroclinic processes as the lower level baroclinic eddy generation and vertical propagation of the eddy activity. In a conservative (i.e., adiabatic and frictionless) flow, the time mean eddy momentum flux convergence is solely determined by the baroclinic eddy generation.

- $-k_{eff}\frac{\partial Q}{\partial Y_e}$ represents the irreversible PV mixing (hereafter referred as barotropic term), which is formulated in a diffusive closure. It manifests the contribution from barotropic processes such as the horizontal vorticity advection, wave breaking, and PV contour filamentation that leads to wave dissipation at small scales. k_{eff} denotes the effective diffusivity of irreversible PV mixing (Nakamura 1996), which is estimated as a residual using equation (6) in this study. In the region where wave breaking occurs, large values of k_{eff} are expected. An example of this is shown in Fig.3b, in which a snapshot of the latitudinal distribution of upper-level k_{eff} on day 1300 in the CTL run is displayed. The effective diffusivity reaches maximum at the jet flanks, where wave breaking, PV filamentation and thus small-scale PV diffusion have intensively occurred, as shown in Fig.3a. The effective diffusivity, however, exhibits a minimum at the jet center, which is consistent with the notion that a strong upper-level jet acts as a mixing barrier (Haynes and Shuckburgh 2000; Chen and Plumb 2014). Hence, it is

the stretching and deformation of PV contours that give rise to the enhanced irreversible PV mixing. Eddy kinetic energy, despite attaining a maximum at the jet center, has a secondary control on the distribution of PV mixing. The changes of irreversible PV mixing can be further decomposed into three components:

$$-\Delta(k_{eff}(Y_e,t)\frac{\partial Q(Y_e,t)}{\partial Y_e}) = -\Delta k_{eff}\cdot\partial Q/\partial Y_e - k_{eff}\cdot\Delta\partial Q/\partial Y_e - \Delta k_{eff}\cdot\Delta\partial Q/\partial Y_e. \quad (7)$$

Thus, the changes of irreversible PV mixing could be attributable to the change in effective diffusivity ($-\Delta k_{eff}\cdot\partial Q/\partial Y_e$), change in the background PV gradient ($-k_{eff}\cdot\Delta\partial Q/\partial Y_e$), and the covariance of the changes ($-\Delta k_{eff}\cdot\Delta\partial Q/\partial Y_e$).

- $-\frac{\partial A}{\partial t}$ is the negative time tendency of wave activity A. In a pure barotropic conservative flow, $A+[u]$ is constant even if nonlinear wave breaking occurs. Then an increase of wave activity indicates a jet deceleration. An increase/decrease of wave activity on the jet's equatorward/poleward flank will lead to a poleward jet shift.

- $\Delta\Sigma$ denotes the diabatic source/sink of wave activity. As in Sun et al. (2013), in this study it is computed by

$$\Delta\Sigma = \frac{1}{L_{ength}}(\iint_{Q(Y_e)\leq q(y), y\leq Y_0} -f_0\frac{\partial}{\partial p}\frac{HR}{spC_p}dS - \iint_{Y_e\leq y\leq Y_0} -f_0\frac{\partial}{\partial p}\frac{HR}{spC_p}dS), \quad (8)$$

which measures the net areal displacement of diabatic heating from the zonal symmetry.

In summary, equation (6) provides a feasible diagnostic framework to separate the roles of baroclinic and barotropic processes in determining the eddy momentum flux convergence. The impacts of lower level thermal forcing on the upper tropospheric wave propagation and breaking, as well as on the lower tropospheric eddy generation and subsequent eddy momentum flux convergence can be evaluated.

We also want to point out that the budget equation (6) actually derives from the E-P flux divergence, thus it is essentially consistent with the conventional Transformed Eulerian Mean (TEM) framework. In the quasi-geostrophic limit,

$$\nabla \cdot F = -\frac{\partial [u^* v^*]}{\partial y} + \frac{\partial}{\partial p}\left(\frac{f[v^* \theta^*]}{\partial \overline{\theta}^{xy}/\partial p}\right) \approx -k_{eff}\frac{\partial Q}{\partial Y_e} - \frac{\partial A}{\partial t} + \Delta\Sigma. \tag{9}$$

The zonal wind acceleration $\frac{\partial [u]}{\partial t} = f[\tilde{v}] + \nabla \cdot F + [F_r]$, in the TEM formalism. However, as shown in Pfeffer (1987), the Coriolis force acting on the residual circulation $[\tilde{v}]$ is always comparable or even larger in magnitude than the E-P flux divergence and of opposite sign. The net zonal wind acceleration is given by a small difference between these two large quantities. The zonal wind acceleration is actually more closely related to the convergence of eddy momentum flux than the E-P flux divergence. This is another reason that, in this study, we choose to adopt the FAWA framework to diagnose the eddy response. In the following sections, we still analyze the response of the E-P flux, mainly as indicators for the variations of the eddy momentum and heat fluxes.

3. Results of Standard Experiments

a. Equilibrated response

The equilibrated atmospheric response to the meridional displacement of the lower-level thermal forcing is presented. Fig.4a displays the climatology of the zonal-mean zonal wind and meridional temperature gradient ($\partial[T]/\partial y$) in the standard control run. The CTL run simulates an eddy-driven jet at the center of the channel, with the strongest baroclinicity peaking at the jet center. The equilibrated response of the zonal-mean zonal winds and temperature gradient in the standard LOW runs compared to the CTL runs is also displayed. As shown in Fig.4b, for the LOW run where $\Delta y = 1250$ km, the zonal wind manifests an equivalent barotropic dipolar response centered at the mid-channel with positive anomalies on the poleward side, indicating a poleward shift of

the jet stream. Consistent with the zonal wind response, the response of temperature gradient in the troposphere is characterized by an equivalent barotropic dipolar structure, denoting a poleward shift of the baroclinic zone.

The sensitivity of the atmospheric response to the imposed lower-level thermal forcing is summarized in Fig. 5. The latitude of the maximum zonally averaged surface wind, upper-tropospheric zonal wind, and barotropic (i.e., vertically averaged) wind is shown as a function of the latitudinal displacement distance Δy of the thermal forcing. This latitude is obtained by estimating the location where the meridional derivative of zonal wind is close to zero near the grid point of the zonal wind maxima using the cubic interpolation. For an increase in the latitudinal displacement of the thermal forcing, the zonal winds show an approximately linear poleward shift. To denote the movement of the baroclinic zone, the latitude of the maximum baroclinicity at the lower tropospheric is also displayed in Fig.5a. Consistent with the poleward displacement of the zonal winds, the lower-level baroclinicity exhibits a monotonic poleward displacement with the poleward shift of the thermal forcing. Meanwhile, the latitude of the maximum eddy kinetic energy (EKE) as a function of Δy in the LOW runs is also displayed in Fig.5b. The upper tropospheric EKE exhibits a monotonic poleward shift to the increase in the latitudinal displacement of the thermal forcing similar to the response of the zonal wind. Thus in the following analysis, only the atmospheric response in the experiments with $\Delta y = 1250km$ is shown.

The roles of eddies in shifting the zonal flow is further investigated through the responses of the zonal-mean eddy statistics. The variation of the zonal mean eddy momentum flux convergence is plotted in Fig.6a. The response patterns of the eddy momentum forcing match very well that of the zonal wind, indicating the importance of the eddy feedback in the jet shift. From the perspective of the thermodynamics, the variation of the eddy heat flux is also an important factor contributing to the poleward shift of the baroclinic zone. Consistent with the variation of the

meridional temperature gradient in Fig.4b, the response of the eddy heat flux in Fig.6a also exhibits a dipolar structure, indicating a poleward shift of the eddy generation.

The equilibrated response of eddy activity is also depicted in the E-P cross section. As shown in Fig.6b there is anomalous upward wave activity flux on the poleward side and downward wave activity flux on the equatorward side. Along with the anomalous wave activity flux, the E-P flux on the poleward side exhibits a divergence near surface as a direct response to the change in lower-level baroclinicity (Ogawa et al. 2012), as well as a convergence in the lower troposphere and a divergence in the upper level. All the above eddy responses indicate that, associated with the poleward jet shift, the baroclinic zone shifts poleward and the eddy activities are generated more vigorously on the poleward side of the jet, which further enhances the poleward jet shift.

b. Transient Response

To address the key dynamical processes responsible for the latitudinal shift of the zonal flow and eddy fluxes, the transient atmospheric response to a "switch-on" thermal forcing is studied. 200-member ensembles of transient experiments with Δy= 1250 km are performed. Note that despite the sample size is 200, the internal noise remains visible in the ensemble mean due to the internal variability of the eddy-zonal flow interactions. The following investigation is based on the ensemble mean features.

Figures 7a-b show the temporal evolutions of the surface and upper tropospheric anomalous zonal winds in the transient experiment. The surface zonal wind displays an initial equatorward shift during the first 27 days and a poleward shift afterwards. The equilibrium is approximately reached after about day 80. The upper tropospheric zonal wind displays a monotonic poleward displacement with a magnitude greater than the surface wind. By comparison, the temporal evolution of the vertically averaged eddy momentum flux convergence is shown in Fig.7c. The response

of the eddy momentum forcing during the initial several days is very weak. After around day 12, it exhibits a poleward shift similar to the zonal winds.

The temporal relationship between the eddy momentum forcing and the zonal wind is analyzed by projecting them onto the equilibrated response pattern of the surface zonal wind in the standard LOW run, which denotes a meridional displacement of the eddy-driven jet. Figure 7d shows the projected time series of the surface zonal wind and the vertically averaged eddy momentum flux convergence. The evolution of the surface zonal wind, as expected, is correlated with eddy momentum flux convergence.

Based on the temporal evolutions of zonal winds and eddy momentum flux convergence in Fig.7d, the transient atmospheric response in the ensemble experiments can be roughly divided into three different stages. In stage I (days 1-12), the zonal wind response displays a baroclinic structure. The surface zonal wind shifts equatorward but upper tropospheric zonal wind shifts poleward. Meanwhile, the response of eddy momentum flux convergence is very weak. During stage II (days 13-27), the surface zonal wind is still at the equatorward side to its initial position, but it begins to move poleward. The eddy momentum flux convergence at this period has a weak poleward shift. During stage III (days 28-80), the surface zonal wind moves to the poleward side of the initial position and continues to shift poleward with a larger magnitude. Both the upper tropospheric zonal wind and barotropic eddy momentum flux convergence display a stronger poleward shift.

Vertical structures of the changes of zonal-mean zonal wind and meridional temperature gradient in the above three time periods are plotted in Figure 8. The zonal wind response in the initial 12 days (Fig.8a) displays a poleward shift with baroclinic vertical structure, consistent with Hoskins and Karoly (1981). As the change in eddy forcing during this initial period is small, this can be thought of as a thermal wind adjustment to the imposed lower-level thermal forcing. The weak and

equatorward shift of the surface wind in this stage as in Fig.7a is mostly driven by the anomalous overturning circulation in response to the diabatic forcing (results not shown here). Following the initial adjustment, the response of zonal wind from day 13 to day 27 (Fig.8b) becomes progressively more barotropic and increases in both spatial extent and magnitude. After day 28, the zonal wind response exhibits an equivalent barotropic vertical structure (Fig.8c) and the amplitude of the response during this period is much stronger than the first two periods.

The causes for the transient responses of the zonal wind and its attendant eddy momentum forcing are explored by examining the temporal evolution of the upper tropospheric FAWA budget. This is analyzed by projecting each forcing term in equation (6) onto the equilibrated response pattern of the surface zonal wind. The obtained time series of each forcing term are shown in Figure 9a. The FAWA budget shows that the barotropic and baroclinic eddy processes contribute differently to the poleward shift of eddy momentum flux convergence. The barotropic term shows a consistent positive contribution to the eddy momentum forcing, suggesting its dominant role in the poleward shift of eddy momentum flux convergence. The baroclinic term, however, exhibits a regime transition. It attempts to shift the zonal wind equatorward in the initial stage of the transient response, which offsets the positive contribution of the barotropic processes, then begins shifting poleward after day 21, and finally contributes positively to the poleward shift of the eddy momentum forcing. The wave activity tendency has a positive contribution initially and becomes zero in the equilibrium state. The contribution of diabatic heating is in general much weaker. Additionally, the dominant barotropic term is further decomposed as in equation (7), whose time series of each component is displayed in Figure 9b. It is clear that the positive contribution of barotropic processes is primarily due to the change in effective diffusivity.

The dynamical mechanism responsible for the poleward shift of the eddy momentum flux convergence is examined by the vertical structures of the E-P flux and FAWA budget. Since the zonal

wind response at stage I is interpreted as a thermal wind adjustment to the lower level thermal forcing, we will focus on the changes during stages II and III, which are supposed to be the key processes in driving the jet shift. Figure 10a illustrates the changes of the E-P flux, its divergence and eddy momentum flux convergence during stage II. Large area over the channel center is dominated by the downward E-P flux, with upward wave activity flux only confined in the lower troposphere in the poleward side of the channel due to the enhancement of the lower-level baroclinicity there. In the upper troposphere, the E-P flux exhibits a divergence in the poleward latitudes and convergence in the equatorward latitudes, which leads to a poleward shift of the eddy momentum flux convergence.

The change of the E-P flux divergence is interpreted through the changes of the FAWA budget in the upper troposphere. As shown in Figs.10b-e, the poleward shift of the eddy momentum forcing during stage II mainly results from the change of the barotropic processes in the upper troposphere. The change of the wave activity tendency appears to be in quadrature with the change of eddy momentum forcing and thus contributes little to the shift. The change of the baroclinic process offsets the poleward shift of the eddy momentum forcing, which is consistent with the results in Fig.9a. In addition, the change of the barotropic processes is further decomposed into the contributions from the change of effective diffusivity and the change due to the background PV gradient. As shown in Figs.10f and g, the change of the barotropic processes is primarily due to the change of the effective diffusivity. This result is also consistent with the time evolution of the FAWA budget in Fig.9b.

Figure 10h displays the vertical structure of the anomalous effective diffusivity during stage II. The effective diffusivity shows a reduction on the poleward side and an enhancement on the equatorward side of the channel. This can be thought of as the result of a strong zonal jet in suppressing the effective diffusivity of PV mixing (Haynes and Shuckburgh 2000; Ferrari and

Nikurashin 2010; Chen and Plumb 2014). As such, less/more wave activity will be dissipated on the jet's poleward/equatorward flank. More vertically propagating baroclinic wave activity fluxes can survive on the jet's poleward flank and further propagate to the equatorward side.

The change of effective diffusivity during stage II is complemented by the statistics of cyclonic wave breaking (CWB) versus anticyclonic wave breaking (AWB) events. Irreversible PV mixing often takes place near the critical latitudes where the wave breaking events occur frequently. Thus it is anticipated that the region of frequent wave breaking coincides with the region of large effective diffusivity. Following the method suggested in Riviere (2009), the wave breaking event is detected by counting the point where the PV contour has a local reversal. Figure 10i shows the change of Rossby wave breaking frequency at 312.5 hPa during stage II. Consistent with the change in effective diffusivity, both the AWB and CWB frequency show a reduction on the poleward flank of the jet and an enhancement on the equatorward side of the jet, with the frequency of the AWB increasing more than the CWB on the equatorward flank. However, the asymmetry of the CWB/AWB variation with the jet shift is far less dramatic than that in Riviere (2009), implying that the jet shift in this stage mostly attributes to the change in the location of the CWB/AWB other than the change in the preferred type of wave breaking as suggested in Riviere (2009).

Similar analyses on the E-P flux and FAWA budget are also applied to stage III. As shown in Fig.11a, the E-P flux shows stronger upward (downward) wave activity flux on the poleward (equatorward) side, giving rise to strong divergence (convergence) at the surface and convergence (divergence) aloft, which pushes the jet further poleward. A comparison between Figs.11a and b-e show that the poleward shift of the eddy momentum flux convergence during the period is mainly attributed to the contribution of baroclinic term. In this stage, the upward eddy activity flux on the poleward side is enhanced, which contributes to the poleward shift of the eddy momentum flux convergence. On the other hand, the barotropic term in Fig.11b shows a quadrapolar structure

change, which is primarily due to the change of background PV gradient as in Fig.11g. The changes of eddy effective diffusivity and Rossby wave breaking during this stage, as shown in Figs.11h and i, display consistent poleward displacement. Their contribution on the barotropic term, as shown in Fig.11f, offsets the contribution from the change of PV gradient (Fig.11g). Thus the total contribution of barotropic processes to the poleward shift of the eddy momentum forcing is relatively weak.

The above analysis suggests that, in addition to the thermal wind adjustment, both the barotropic and baroclinic processes play a role in the eddy feedbacks to the low-level thermal forcing. The key processes that lead up to the poleward shift of the eddy-driven jet are summarized as follows.

- **Stage I (day 1-12):** With an abrupt poleward shift of the low-level thermal forcing, the zonal wind adjusts almost instantaneously to the thermal wind relationship, leading to an enhancement of the upper-level zonal wind on the poleward side and a reduction of zonal wind on the equatorward side.

- **Stage II (day 13-27):** The enhanced upper-level zonal wind on the poleward side reduces the local effective diffusivity. As a result, more vertically propagating baroclinic wave activity fluxes can survive the local dissipation on the jet's poleward side and propagate to the equatorward side. The resultant anomalous eddy momentum flux converged on the poleward side and further pushes the jet shifting poleward.

- **Stage III (day 28-80):** Along with the jet shift, lower level baroclinic eddies tend to generate more vigorously and propagate upward on the poleward side, which enhances the anomalous eddy momentum flux convergence and pushes the jet further poleward.

4. Design of the Overriding Experiments

The analysis in section 3 provides additional support to recent studies (Chen et al. 2013; Sun et al. 2013; Lu et al. 2013) on the importance of irreversible PV mixing for the jet shift in response to thermal forcing. However, this can not separate the barotropic response from the baroclinic response unambiguously, as barotropic and baroclinic mechanisms are often coupled. To further evaluate the relative importance of the barotropic and baroclinic eddy feedbacks, an overriding technique is used to isolate the effect of barotropic advecting flow on PV mixing. This is motivated by Ferrari and Nikurashin (2010) who separated the zonal flow from eddies in the advection-diffusion equation of passive tracers, and who showed clearly a zonal jet suppresses the effective diffusivity at the jet center in spite of large eddy kinetic energy.

It is well-known that the variability of equivalent barotropic zonal wind anomaly is critical for eddy feedback. In the linear conservative limit, the zonal wind determines the critical latitude of a given wave perturbation. Idealized baroclinic eddy life cycle calculations have indicated large sensitivities of the type of wave breaking to the meridional shear of barotropic zonal wind (e.g. Thorncroft et al. 1993). Chen et al. (2007) showed that surface friction can alter the strength of barotropic flow and thus the latitude of surface westerlies. Kug and Jin (2009) and Ren et al. (2009) demonstrated the importance of barotropic zonal wind in shaping the synoptic eddy feedback by altering the vorticity deformation. A recent study by Nie et al. (2014) also found that the anomalous zonal wind in the Southern Annular Mode variability leads the change in the irreversible PV mixing as well as the frequency distribution of wave breaking. In comparison with other perturbation methods, overriding the barotropic zonal flow is desirable for this study. As shown in Hartmann and Zuercher (1998), the barotropic zonal flow shows slight influence on the baroclinic energy conversion but evidently affects the barotropic decay process in an eddy life

cycle. Thus, overriding the barotropic zonal flow mainly impacts the effective diffusivity by modulating vorticity advection and deformation.

More specifically, if we decompose each variable into the zonal average and the eddy components, the PV advection term in equation (1) can be rewritten as

$$\mathbf{V} \cdot \nabla_h q = [u]\frac{\partial q^*}{\partial x} + v^* \frac{\partial [q]}{\partial y} + \mathbf{V}^* \cdot \nabla_h q^* \tag{10}$$

where * denotes the eddy component and $[\quad]$ denotes zonal average. The first term on the right hand side of the equation just represents the PV advection by the zonal-mean zonal wind. The zonal-mean zonal wind can be further decomposed into the barotropic component of the zonal wind $[u]_{bt}$, which is estimated as the vertical average of the zonal wind, and the baroclinic component of the zonal wind $[u]_{bc}$, which is defined as $[u]_{bc} = [u] - [u]_{bt}$. Thus this term can be further decomposed as

$$[u]\frac{\partial q^*}{\partial x} = ([u]_{bt} + [u]_{bc})\frac{\partial q^*}{\partial x}. \tag{11}$$

In the overriding experiment, we specify the barotropic zonal-mean zonal wind $[u]_{bt}$ in the PV advection and leave all the other terms, e.g. baroclinic zonal wind in the PV advection, and all the other fields, e.g. wind, temperature and wave activity, determined by the model evolution. As such, the overriding technique decouples the barotropic PV advection from baroclinic energy production. By doing so, we aim to suppress the barotropic eddy feedback associated with the variability in barotropic zonal wind, but to keep the baroclinic eddy feedback associated with the variability in meridional temperature advection and baroclinic zonal wind.

5. Overriding Forced Experiments: Barotropic versus Baroclinic Feedbacks

In this section, the relative contributions of baroclinic and barotropic processes to the latitudinal shift of the eddy-driven jet will be separated and evaluated through a group of overriding

experiments. More specifically, by overriding the barotropic zonal wind in the PV advection and specifying the lower-level thermal forcing, two experiments, which are baroclinic / barotropic response runs, are carried out to quantify the baroclinic / barotropic eddy feedbacks, respectively.

- **The baroclinic response run** is designed to keep the baroclinic eddy feedback active but suppress the barotropic eddy feedback. In the experiment, the lower-level thermal forcing is kept the same as in the standard LOW run, but the barotropic zonal wind $[u]_{bt}$ in the PV advection is fixed as the climatological mean $[u]_{bt}$ in the standard CTL run. By doing so, the effects of lower-level baroclinicity on the baroclinic eddy generation thus the baroclinic eddy feedback are all kept active, but the transient $[u]_{bt}$ solved by the model is not allowed to affect the PV advection thus the barotropic eddy feedback is suppressed.

- **The barotropic response run** is designed to keep the barotropic eddy feedback but remove the baroclinic eddy feedback. In the experiment, the thermal forcing in the CTL run is used, while the climatological mean $[u]_{bt}$ in the standard LOW run is used in the PV advection. By doing so, no latitudinal shift of the lower-level baroclinicity with the thermal forcing is added into the model, the anomalous baroclinic eddy generation in the LOW run is thus removed. However, as the $[u]_{bt}$ in the LOW run is specified in the PV advection, the influence of the anomalous zonal wind on the irreversible PV mixing is mostly maintained, the barotropic eddy feedback in the LOW run is thus kept in the experiment.

In addition to the above two experiments, another overriding experiment is also carried out for comparison, in which the thermal forcing in the LOW run is used and the $[u]_{bt}$ in the PV advection is overridden by the time mean $[u]_{bt}$ in the LOW run as well. In the experiment, both the baroclinic and barotropic response to the lower-level thermal forcing are expected to be included. By

comparing the results in the overriding runs with the standard LOW run, the relative importance of baroclinic and barotropic eddy feedbacks can be explicitly estimated.

Figs.12a-d show the zonal wind response in the standard LOW run, baroclinic / barotropic response runs and the overriding run including both baroclinic and barotropic response, respectively. Note that in the overriding experiments, we only fix the barotropic flow in the PV advection, thus the zonal wind response is still determined by the model evolution. As shown in Fig.12b, the zonal wind response in the baroclinic response run exhibits a baroclinic vertical structure, denoting a weak poleward shift of the upper-level jet. The pattern of the zonal wind response in the experiment is similar to the initial thermal wind adjustment in the first few days of the transient response, as shown in Fig.8a. In contrast, the zonal wind response in the barotropic response run (Fig.12c) is much stronger and displays an equivalent barotropic dipolar structure, denoting an evident poleward shift of the eddy-driven jet. The zonal wind response in Fig.12c is close to the total zonal wind response in the LOW run (Fig.12a), which indicates the dominant role of barotropic eddy feedback in driving the latitudinal shift of the eddy-driven jet in response to the lower-level thermal forcing. The zonal wind response in the overriding run including both baroclinic and barotropic responses in Fig.12d displays a more similar pattern to the equilibrated response in the standard LOW run.

Similar results are also found for the baroclinic component of the zonal wind response, which is estimated as the difference between the total zonal wind and the vertically averaged zonal wind. As shown in Figure 13b, the response of the baroclinic zonal wind in the baroclinic response run displays a vertical structure very different from that in the standard LOW run in Fig.13a. The baroclinic zonal wind response in Figs.13c and d shows a similar pattern to Fig.13a, characterized by strong vertical wind shear.

The above different characteristics of the zonal wind responses can be further explained from the plots of the E-P flux vector, E-P flux divergence and eddy momentum flux convergence in Fig.14. In the baroclinic response run, as shown in Fig.14b, the E-P vector shows pronounced upward wave activity flux on the poleward side and downward wave activity flux on the equatorward side, which is primarily induced by the poleward shift of the lower level baroclinicity. The upward/downward wave activity flux gives rise to the E-P flux convergence/divergence in the middle troposphere, which is consistent with the response of the mid-tropospheric E-P flux in the LOW run in Fig.14a and implies that our baroclinic response run can well simulate the anomalous baroclinic eddy generation in response to the thermal forcing. However, the eddy response in the experiment is mostly confined in the middle to lower troposphere. The anomalous lower level baroclinic eddies cannot further affect the upper troposphere, thus the response of the upper-level E-P flux convergence and eddy momentum forcing is weak. The anomalous upward/downward wave activity flux mainly acts to reduce/enhance the vertical shear of the zonal wind, as shown in Figs.12b and 13b.

In contrast, in the barotropic response run, as shown in Fig.14c, the E-P flux in the upper troposphere shows strong equatorward propagation response. This induces strong E-P flux divergence/convergence on the poleward/equatorward side, which is most responsible for the dipolar response of the upper-level eddy momentum flux convergence in the standard LOW run. This also shows the validity of our barotropic response run in simulating the barotropic eddy response to the imposed thermal forcing. With the upper-level dipolar E-P flux divergence, an upward/downward wave activity flux is also observed on the poleward/equatorward side. This might result from the meridional overturning circulation and the baroclinic anomalies driven by the upper-level eddy momentum flux convergence, as shown in Nie et al. (2013). However, the resultant E-P flux divergence is much smaller in the lower troposphere than that in Fig.14b, which is consistent with

the strong vertical shear of the zonal wind in Fig.13c. The E-P vector and E-P flux divergence in Fig.14d are mostly the superimposition of the eddy responses in Figs.14 b and c, which are almost the same to that of the total eddy response in Fig.14a. The above comparisons of the zonal wind and eddy responses demonstrate that the barotropic eddy response due to the change in barotropic zonal flow dominates the total atmospheric response to the lower-level thermal forcing.

The effectivity of the overriding technique in delineating the barotropic eddy feedback is also confirmed by comparing the responses of the effective diffusivity between the experiments. As shown in Fig.15b, the response of the effective diffusivity above 400 hPa in the baroclinic response run is in general very weak compared to that in the standard LOW run in Fig.15a, indicating that the barotropic eddy feedback is mostly removed in the baroclinic response run. In contrast, as shown in Fig.15c, the effective diffusivity in the barotropic response run shows evident variations, similar to that in the standard LOW run. Compared to its latitudinal distribution in Fig.3, the effective diffusivity exhibits a poleward shift on both side of the jet, which is consistent the anomalous poleward shift of the zonal wind. This further suggests that the horizontal wave propagation and breaking in the upper troposphere is strongly controlled by the barotropic zonal wind. The barotropic and baroclinic processes can be mostly represented and separated by setting up the overriding experiments.

6. Summary and Discussion

Using a nonlinear multi-level β plane quasi-geostrophic channel model, this study examines the dynamical mechanisms responsible for the latitudinal shift of the eddy-driven jet in response to the meridional shift of the lower-level baroclinicity. Through a FAWA diagnosis and a new overriding technique, the individual roles of the direct thermal wind response, baroclinic and barotropic eddy feedbacks in the jet shift are delineated.

Figure 16 summarizes the mechanisms of the circulation response to the lower-level thermal forcing. First, following a poleward shift of the lower-level baroclinicity, the zonal wind adjusts almost instantaneously by the thermal wind relationship. The upper-level thermal wind adjustment affects the upper-level barotropic processes rapidly: the enhancement/weakening of upper-level zonal wind on the poleward/equatorward side weakens/enhances the local wave breaking and irreversible PV mixing by setting a stronger/weaker barrier of eddy mixing. As a result, there are more baroclinic eddies surviving the local dissipation on the jet's poleward flank and propagating equatorward. The anomalous eddy momentum flux is then converged on the jet's poleward side, which pushes the jet poleward. The baroclinic eddy feedback, however, acts to enhance the poleward shift of the eddy-driven jet only in the later stage of the atmospheric response. Following the poleward shift of the lower-level baroclinicity, baroclinic eddies are generated more vigorously where the baroclinicity increases. The net effect of eddy feedbacks differs from the conventional baroclinic view that the eddy-driven jet responds to the lower-level thermal forcing mostly through baroclinic eddy generation (Hoskins and Karoly 1981). The importance of barotropic eddy feedback is emphasized in our study, which also supports the studies of the circulation response under idealized climate change identified in a dry atmospheric dynamical core (Sun et al. 2013) or an aqua-planet model (Chen et al. 2013).

Furthermore, we have managed to quantify the relative roles of the baroclinic and barotropic eddy feedbacks in driving the latitudinal shift of the eddy-driven jet. By overriding the barotropic zonal-mean zonal wind in the PV advection in our QG model, the barotropic advection and deformation of PV are decoupled from the baroclinic energy conversion, and thus the barotropic and baroclinic eddy feedbacks in the atmospheric response can be delineated. While a number of studies have suggested the importance of the linear wave propagation (Chen et al. 2008; Chen and Zurita-Gotor 2008; Harnik et al. 2010; Barnes 2010) and nonlinear wave breaking (Riviere 2009;

Wang and Magnusdottir 2011; Chen et al. 2013; Lu et al. 2013; Nie et al. 2014) in the upper troposphere, the overriding experiments have demonstrated explicitly that the barotropic response due to the change in barotropic zonal flow dominates the total atmospheric response to the lower-level thermal forcing. Similar to the effective diffusivity of a passive tracer (Haynes and Shuckburgh 2000; Ferrari and Nikurashin 2010; Chen and Plumb 2014), a strong zonal jet suppresses the effective diffusivity in PV mixing in spite of the large baroclinic eddy generation and kinetic energy at the jet center. As such, this overriding experiment corroborates the barotropic eddy feedback mechanism for the persistence of the Southern Annular Mode in the observations (Nie et al. 2014). The overriding experiment also indicates the important role of the barotropic shear in the vertical eddy energy propagation into the upper troposphere. Without the latitudinal shift of the barotropic zonal wind, the anomalous baroclinic energy conversion is mostly confined in the lower levels and cannot efficiently affect the upper-level eddy momentum flux convergence.

The overriding experiment further implies that, in addition to the conventional 'baroclinic adjustment' (Nakamura et al. 2008; Sampe et al. 2010; Zhang and Stone 2011) in response to the extratropical SST anomalies, the barotropic eddy feedback also plays significant roles in the atmospheric response to the lower-level thermal forcing. The change in the upper-level irreversible PV mixing might be most responsible for the anomalous eddy momentum flux as in Sampe et al. (2010) in response to the SST anomalies. The mechanism revealed in this study also provides a dynamical interpretation for the change in Rossby wave breaking frequency during El Nino (Wang and Magnusdottir 2011) or under climate warming (Rivière 2011).

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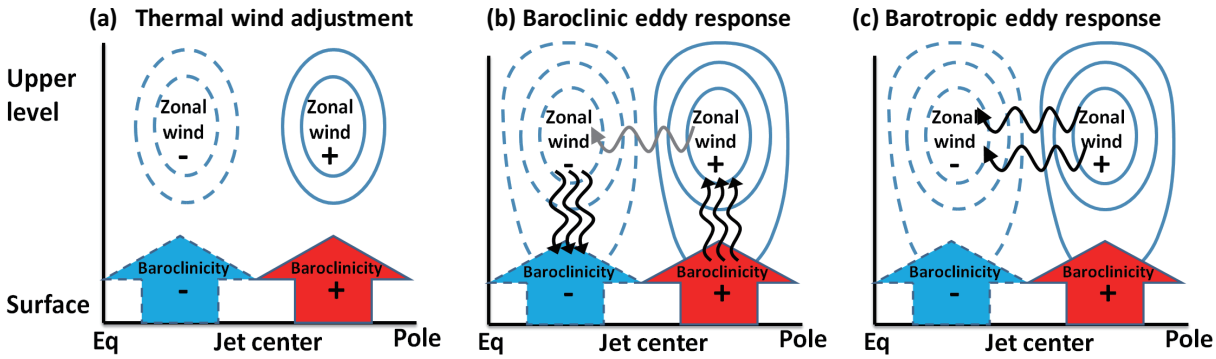

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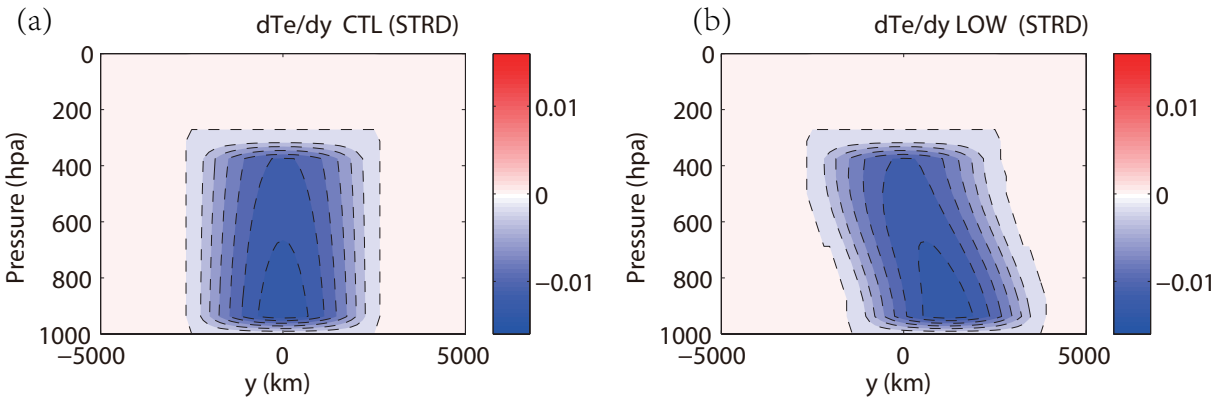

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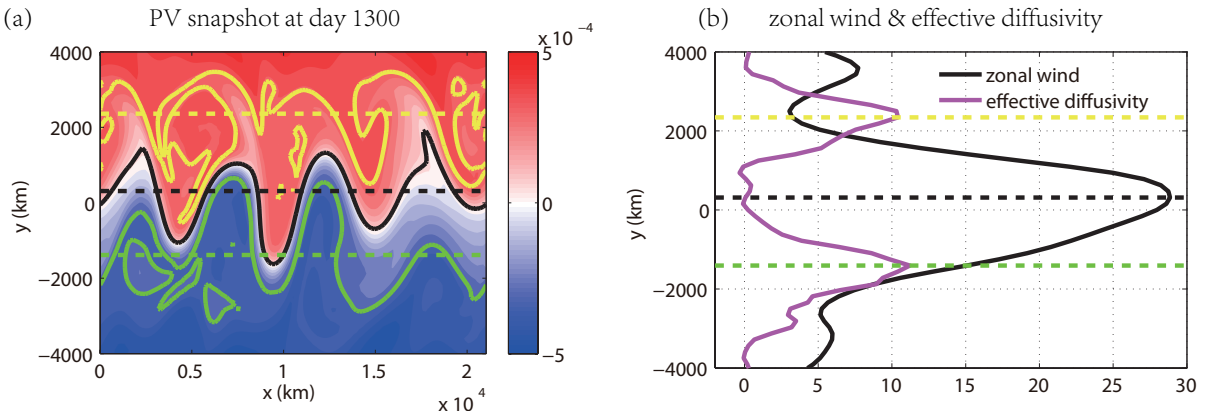

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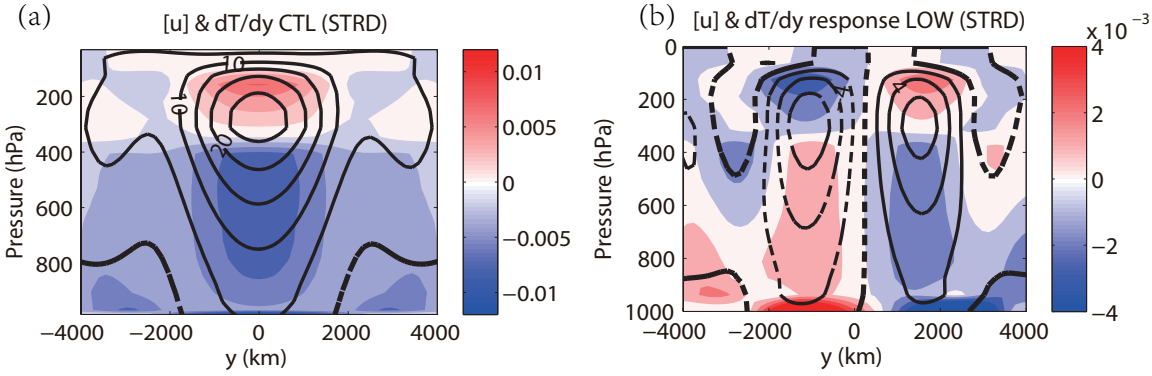

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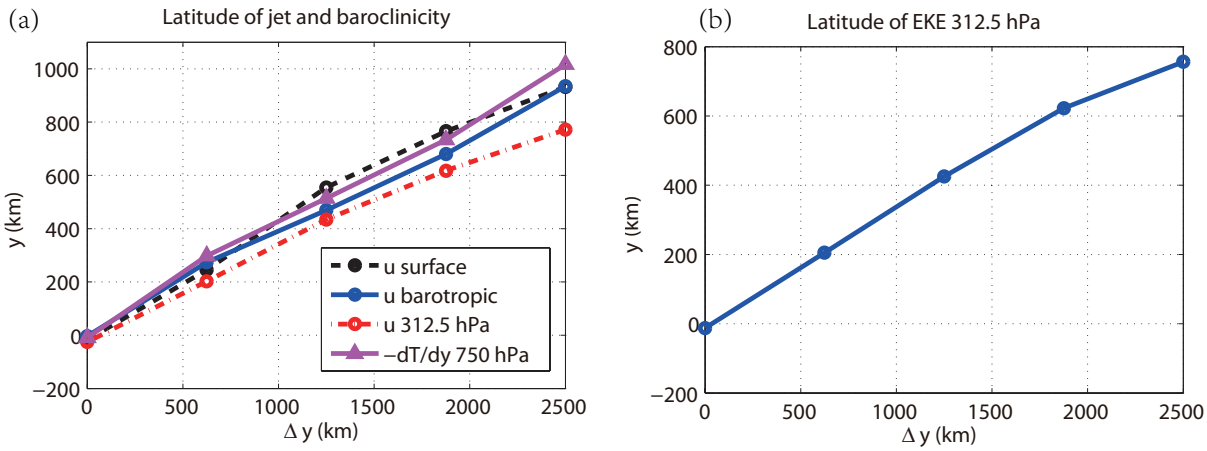

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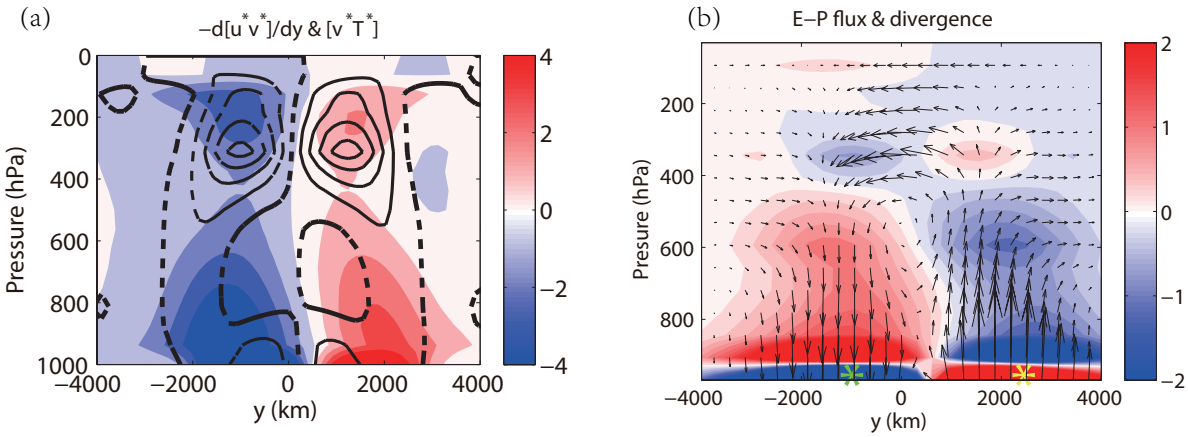

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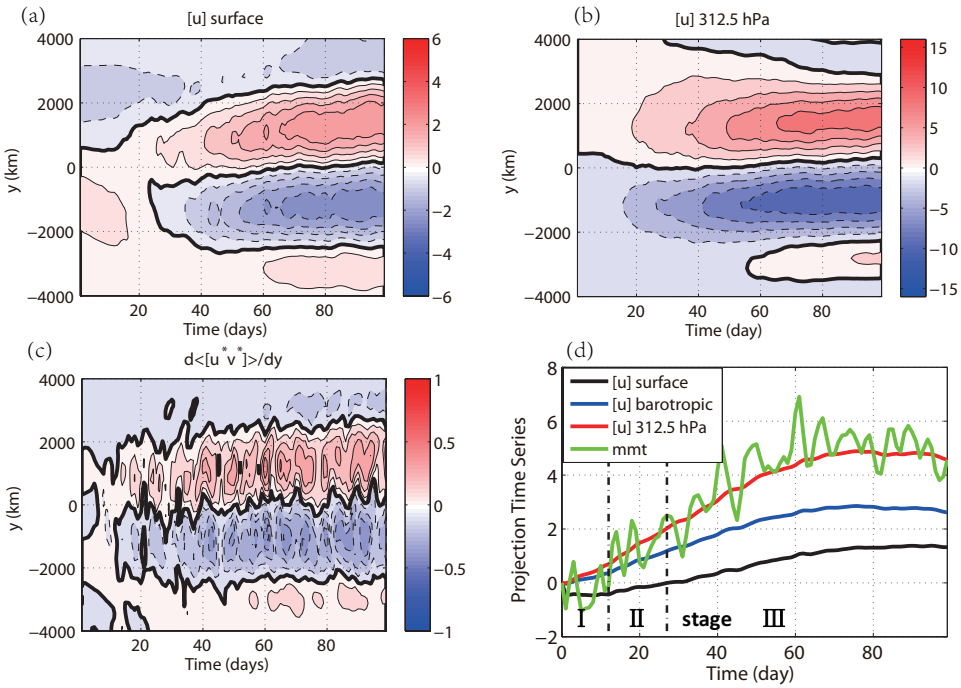

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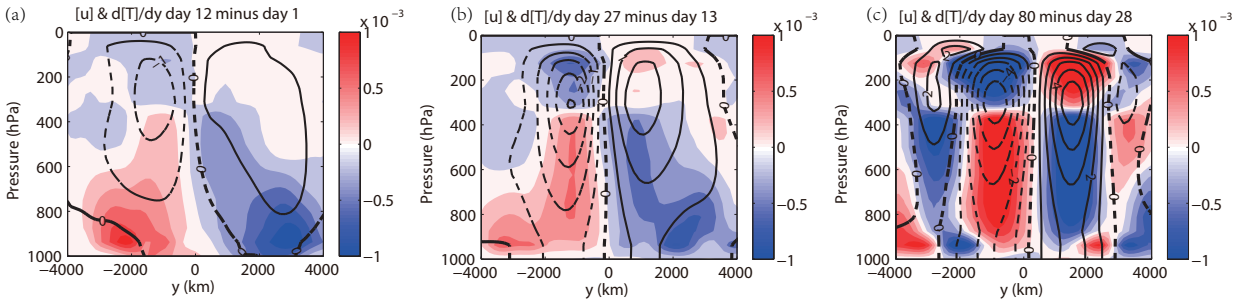

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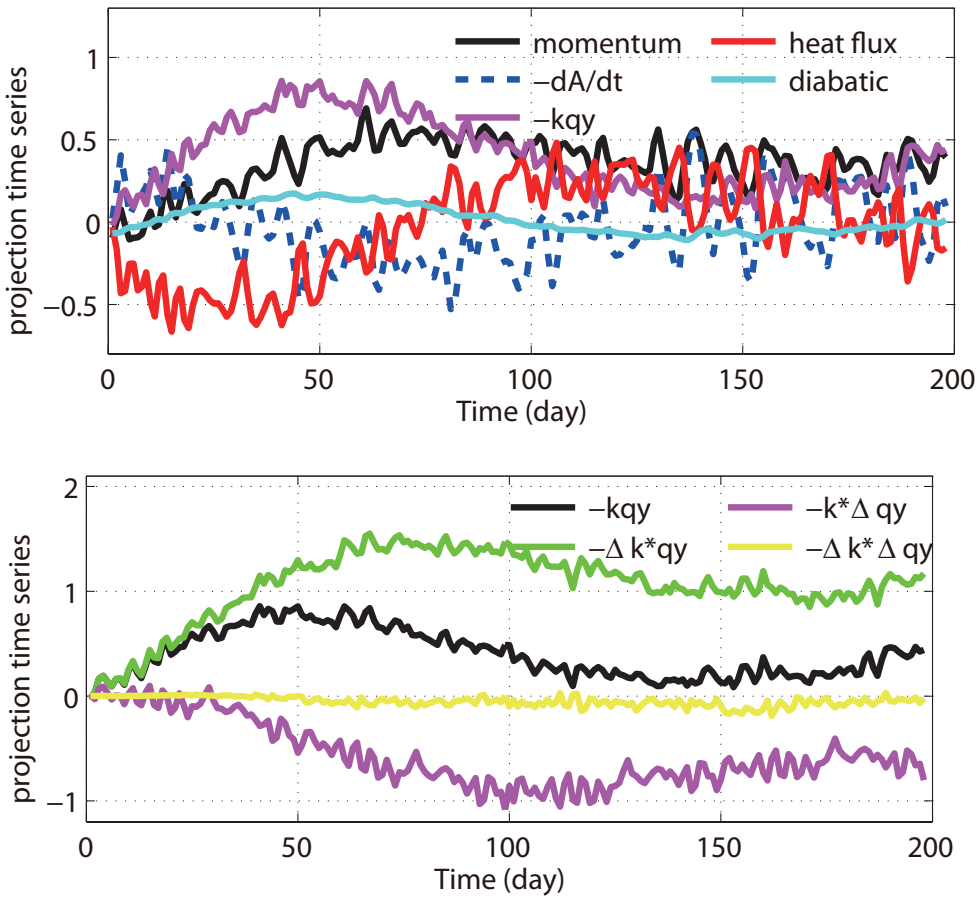

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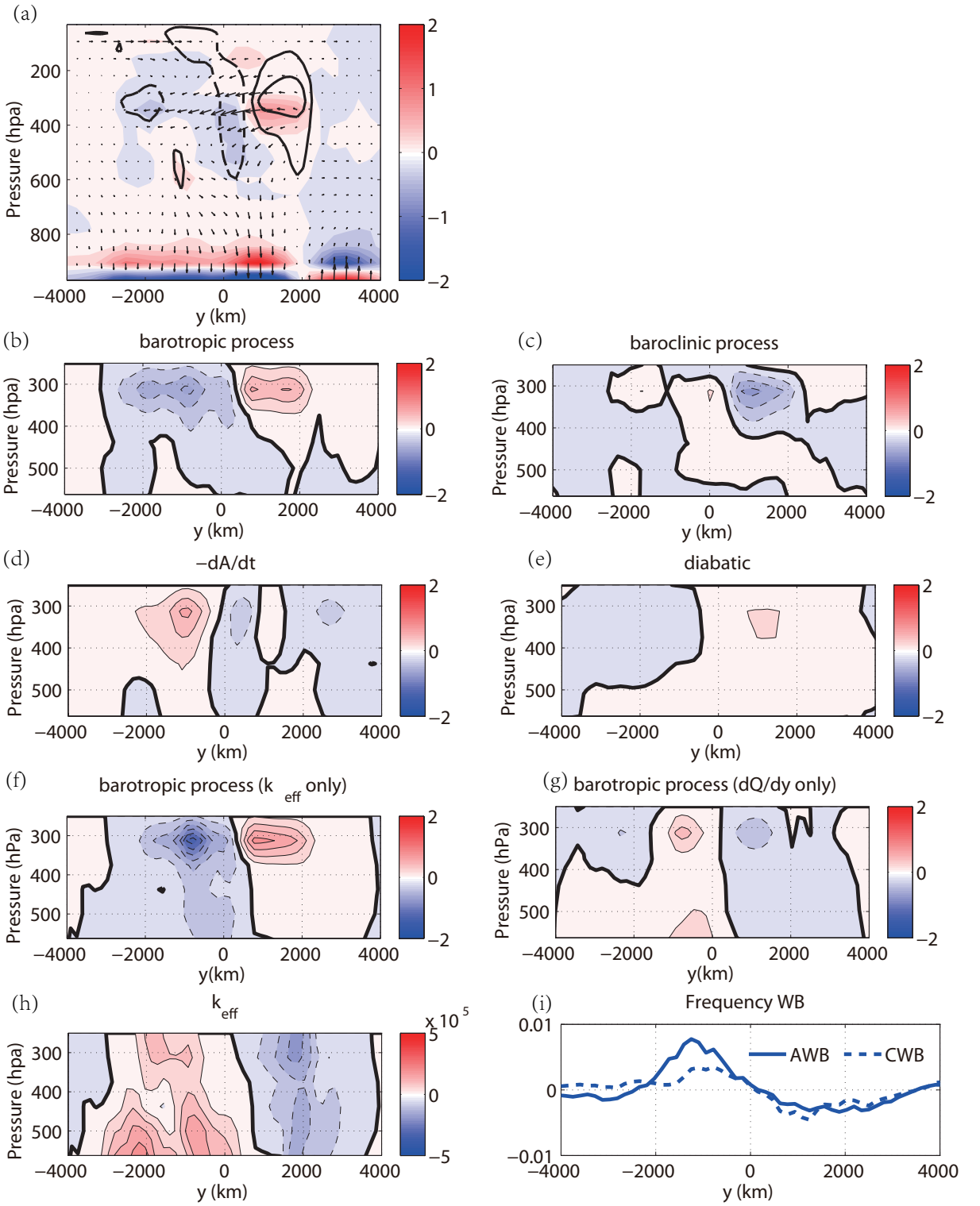

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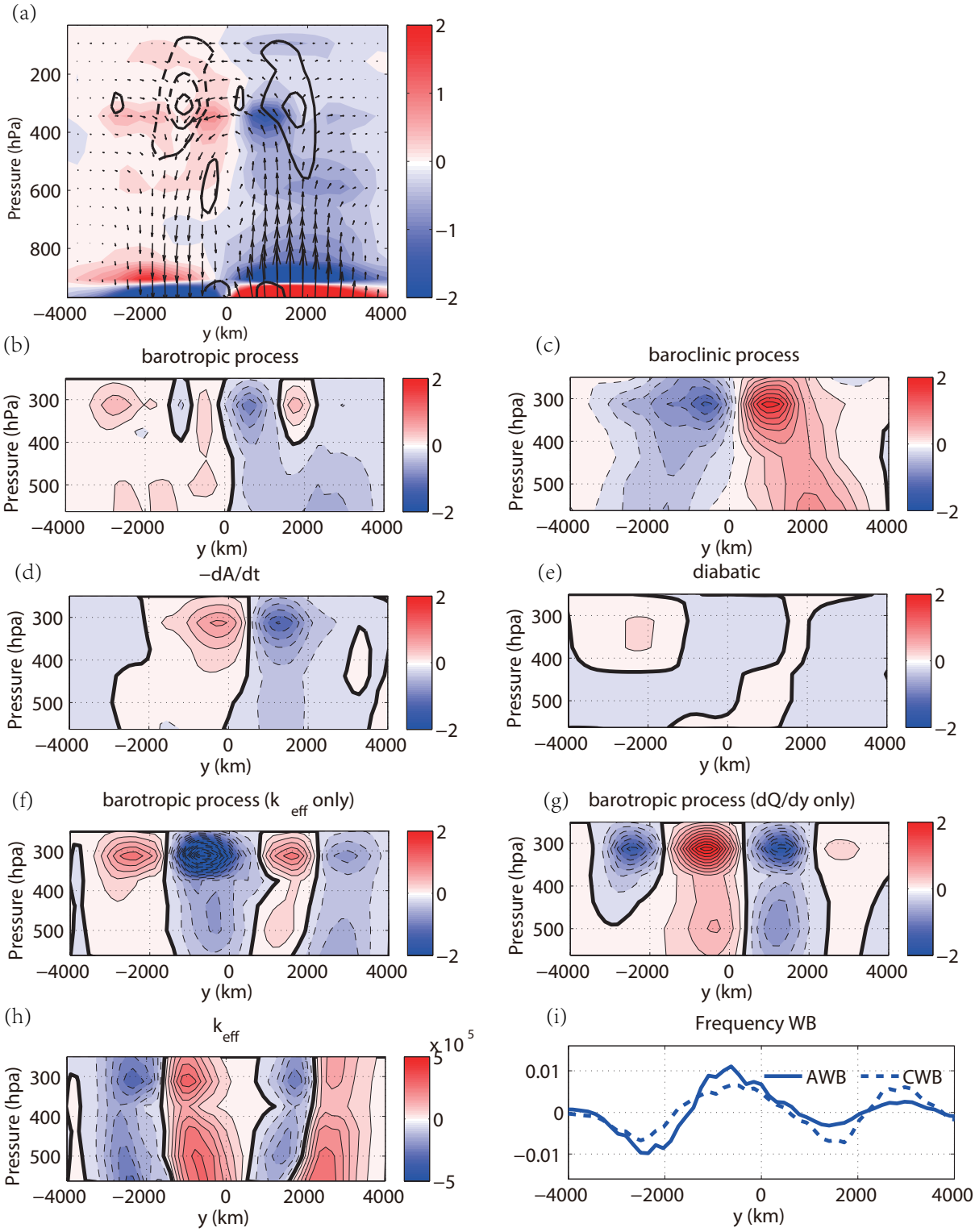

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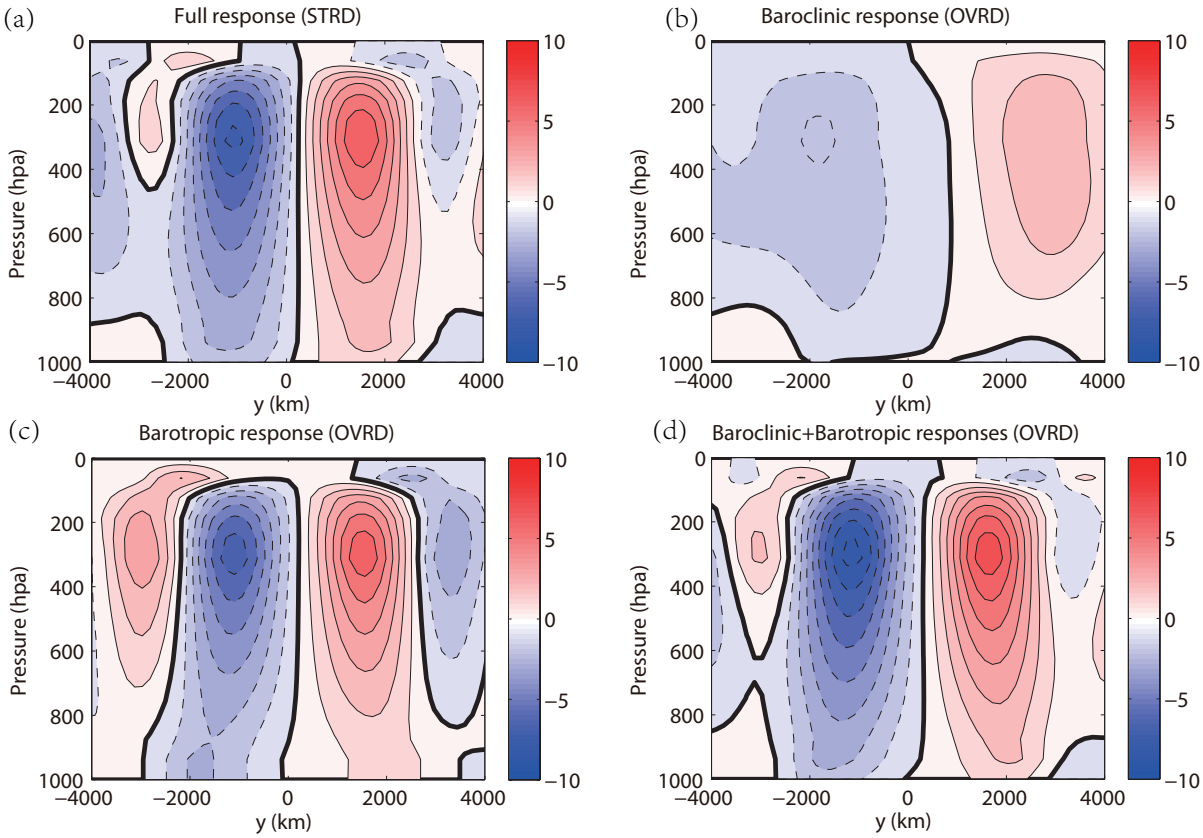

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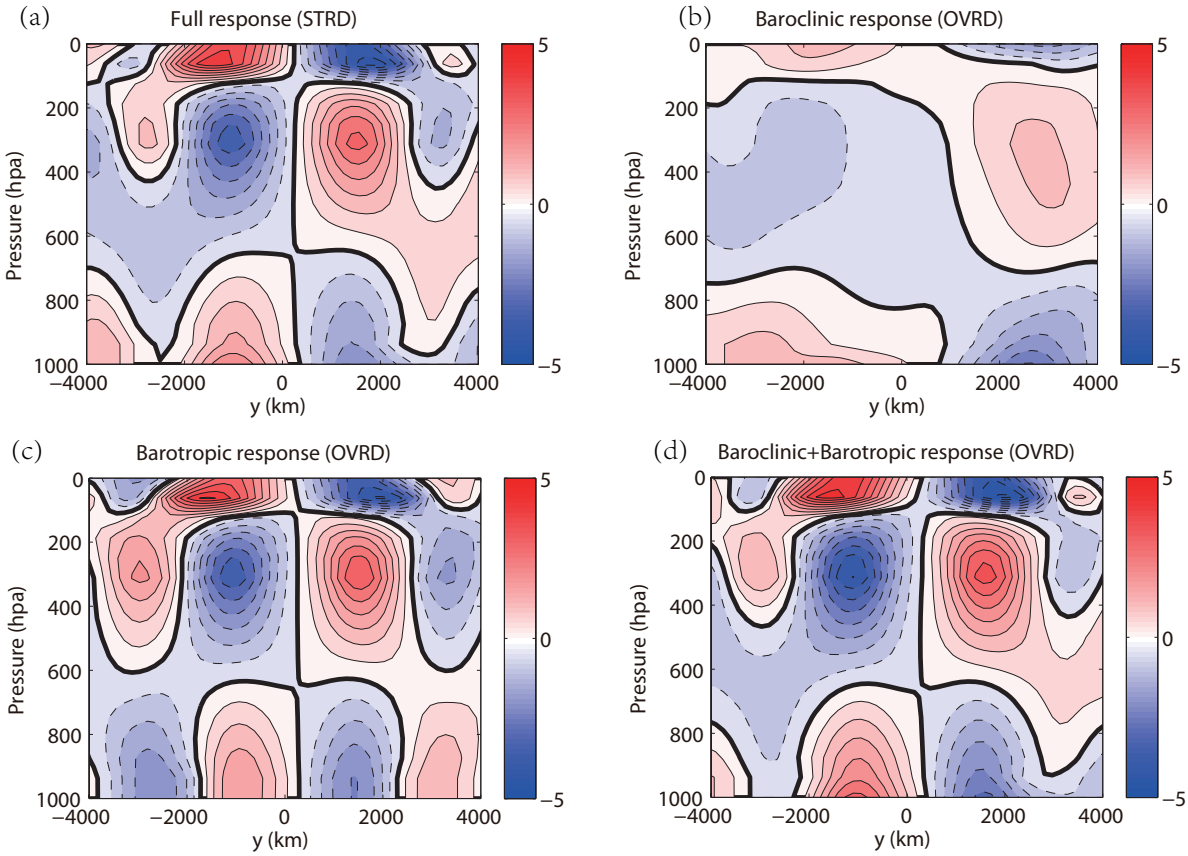

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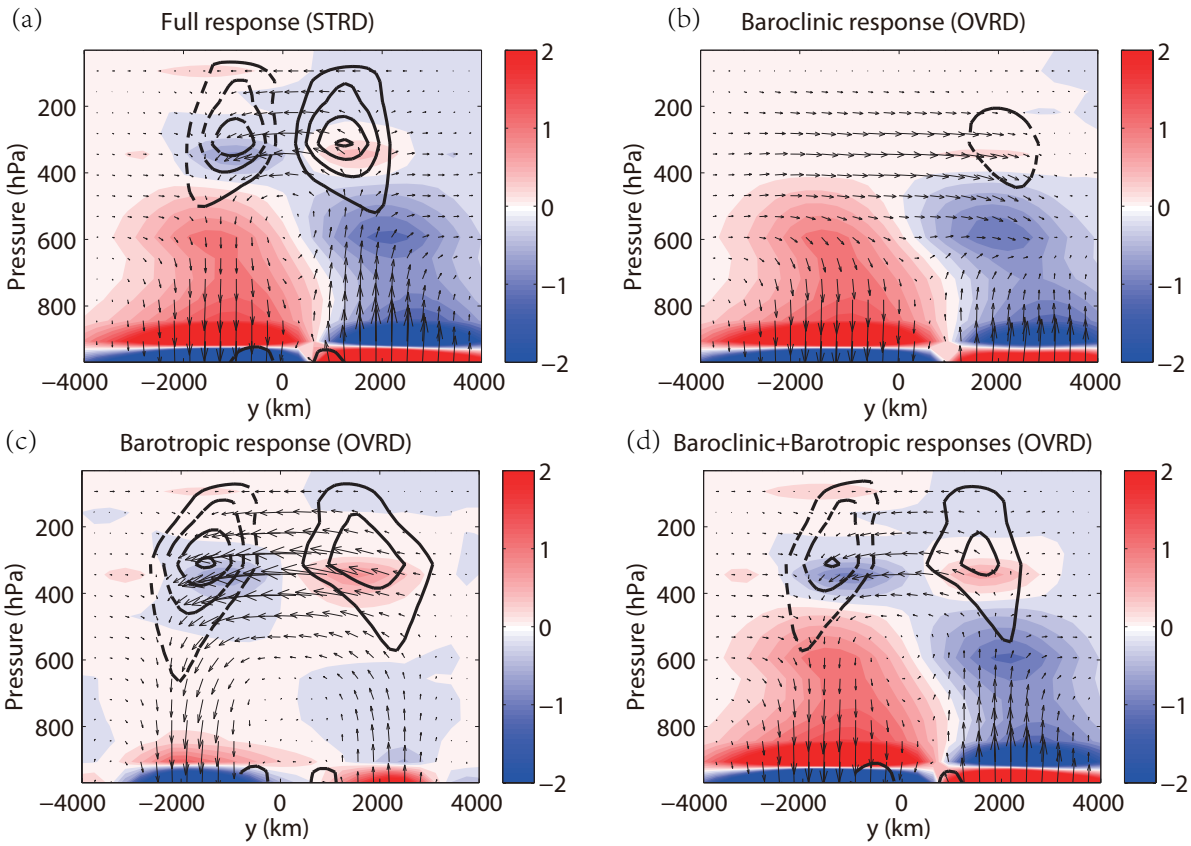

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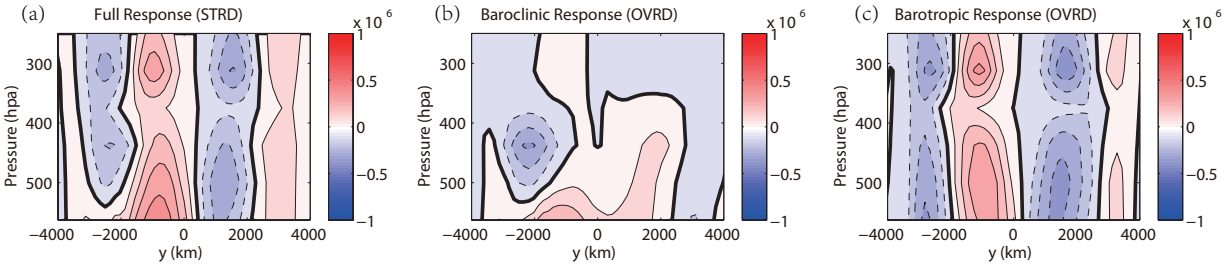

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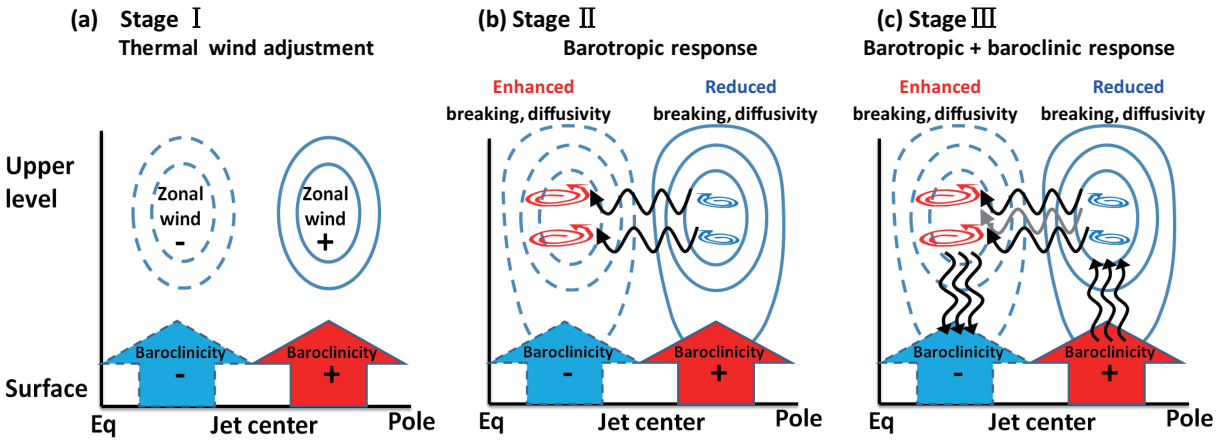

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